

OrbiFarm®

SERIES A · 2026

The Operating System for *indoor* agriculture.

A patented TCEA platform delivering 120x yield, -95% CAPEX, and 85%+ licensing margins — validated by Fraunhofer, protected worldwide.

STAGE

Series A

INSTRUMENT

Equity · incl. tokenize.it

HQ

Ludwigsfelde, DE

CATEGORY

AgTech · Deep Tech



OrbiFarm
Loop Your Food

THE PROBLEM

Agriculture is running out of **land, water, and time.**

01

Shrinking arable land

Soil degradation and climate volatility shrink viable farmland every year — while demand keeps rising.

–33%

Global topsoil lost since 1970

02

Indoor farming is broken

Vertical farms collapsed under crushing CAPEX, energy costs, and labor — InFarm, AeroFarms, Bowery.

>€50M

Typical CAPEX per facility

03

High-value crops stuck offshore

Medicinal plants, APIs, cosmetic actives — grown in unstable regions with zero supply-chain control.

€20B+

Import-dependent plant actives market

MARKET OPPORTUNITY

A **€20B+** controlled-environment market, growing double-digit.

€20B TAM

Global controlled-environment agriculture (CEA) — compounding at **12–15% CAGR** through 2032. OrbiFarm sits at the intersection of three accelerating tailwinds: food security, pharma re-shoring, and sustainability mandates.

12–15% CAGR 2024–2032	3x Verticals Pharma · Food · Cosmetics
€6.5B Medicinal plants SAM by 2030	22+ Species protocols in-house
SOURCE: FORTUNE BUSINESS INSIGHTS, GRANDVIEW · 2025	

THE SOLUTION

OrbiFarm® Loop — a conveyor that **moves the plants, not the lights.**

01 — GRAVITROPISM

Plants root toward gravity

Continuously rotating plant-holders trigger a stress response that boosts biomass and active-compound density.

02 — LIGHT EFFICIENCY

One lamp feeds thousands

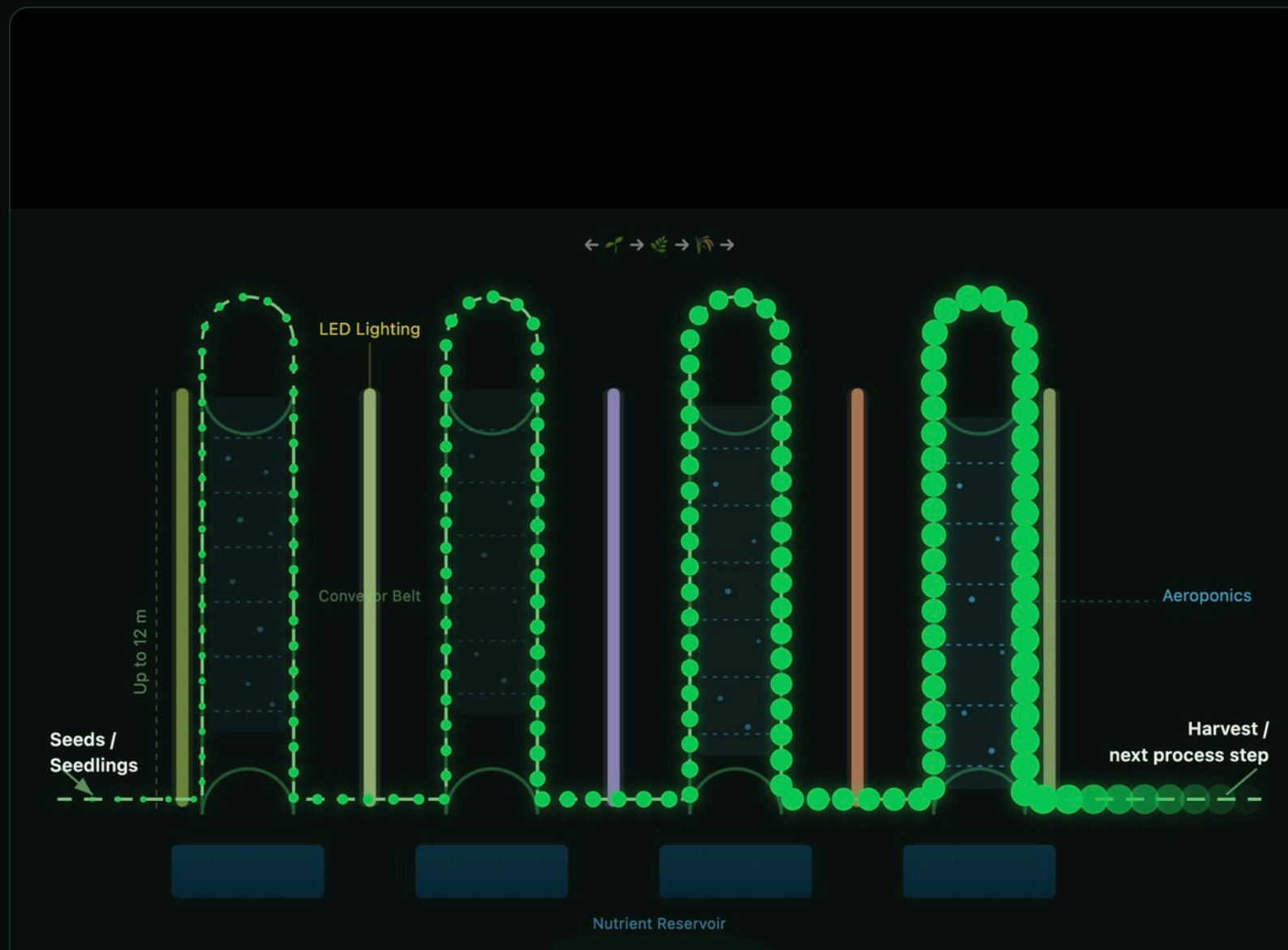
Fixed high-intensity lights; plants cycle past them on a conveyor. Dramatically lower energy per kg of yield.

03 — MODULAR

Container to gigafarm

Stack Loops inside a warehouse or a shipping container — identical unit economics at every scale.

PATENT: EP 3 XXX XXX • WO 20XX/XXXXXX



Numbers that rewrite the unit economics.

120× Yield vs. outdoor per m² footprint	−95% CAPEX vs. indoor vs. conventional vertical farms	−95% Water usage closed-loop system	GMP Pharma-grade ready clean, deterministic output
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01 · ENERGY

Continuous motion = fixed lamps at peak efficiency. **70% lower kWh/kg** than static vertical farming.

02 · THROUGHPUT

Conveyor architecture enables 365-day harvesting with deterministic cycle times — minutes of downtime, not days.

03 · OUTPUT PURITY

Zero pesticides · zero herbicides · zero field contamination. GMP-traceable from seed to extract.

THE MOAT

Patents, protocols, and a three-year head start.

01 · IP

Worldwide patents granted

Core conveyor + gravitropism mechanism protected across EU, US, and key APAC jurisdictions.

02 · KNOW-HOW

22+ plant protocols

Proprietary grow recipes — each a non-replicable licensing asset.

03 · PARTNERS

Fraunhofer-validated

Joint R&D with Fraunhofer IME + Theilen Maschinenbau & Flottweg.



3

Pilot systems live

3+

Years ahead

∞

Scaling potential

TRACTION

Proven in production, not in pitch decks.

**3 pilot systems in continuous operation**

Deployed at Fraunhofer IME — live yield & energy data since 2024.

**Worldwide patents granted & secured**

Core technology IP filed in EU, US, JP, CN — freedom-to-operate confirmed.

**50× higher margins than commodity crops**

Validated on medicinal & specialty plants — priced per gram, not per tonne.

**3 strategic partners signed**

Fraunhofer IME · Theilen Maschinenbau · Flottweg

**22+ plant protocols in-house**

Ready-to-license grow recipes across pharma, food and cosmetics.

**Break-even line of sight FY28**

Clear path to profitability within 3 years on current roadmap.

BUSINESS MODEL

Three revenue streams. **One platform.**

01 · VOLUME

Hardware Sales

OrbiFarm® Loop systems — modular production units sold per site. Drives volume and market penetration.

~25%

Gross margin · upfront revenue

02 · RECURRING

Service & Maintenance

Multi-year service contracts: calibration, spare parts, process optimisation. Predictable revenue per installed unit.

~50%

Gross margin · annual

03 · HIGH MARGIN

Licensing & Royalties

Grow-recipe IP licenses, per-kilogram royalties. Scales with every deployed system — worldwide.

85%+

Gross margin · recurring IP

TARGET MARKETS

One platform, three industries.

01 · PRIORITY

Pharma & Nutraceuticals

€9.2B

Medicinal plants, APIs, cannabinoids. Highest price-per-gram, GMP-ready output — our launch vertical.

Artemisia · Salvia · Cannabis
Ginkgo · Hypericum · Echinacea

02 · SCALE

Specialty Food

€6.8B

High-value herbs, microgreens, edible flowers for HORECA & premium retail. Year-round local supply.

Basil · Cilantro · Saffron
Wasabi · Lemon balm · Chervil

03 · EXPANSION

Cosmetics & Biotech

€4.1B

Bio-active ingredients, fragrance compounds, algal & moss extracts for clean-beauty & biotech.

Rosa damascena · Centella
Calendula · Sphagnum

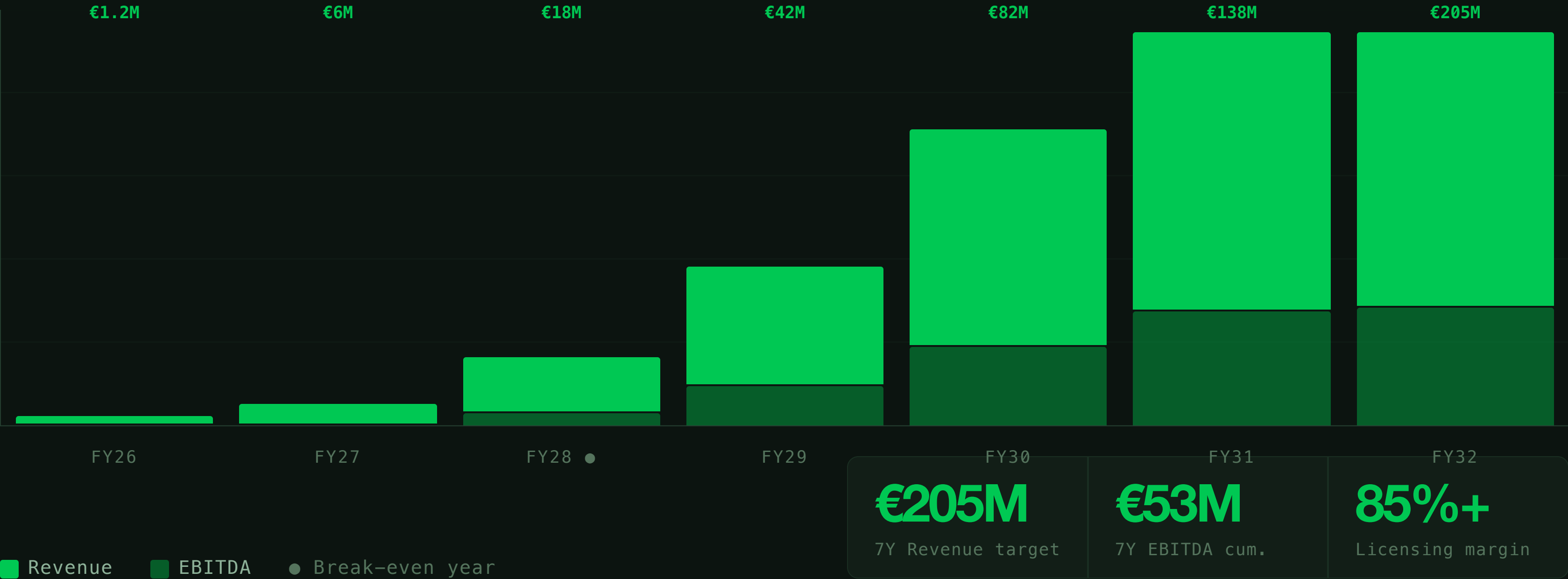
COMPETITIVE LANDSCAPE

Why we win, row by row.

METRIC	ORBIFARM® LOOP	CONVENTIONAL INDOOR / VERTICAL	OUTDOOR / FIELD
CAPEX per m²	Up to 95% lower	€8–15k / m²	Low (but no control)
Yield vs. outdoor	120×	30–50×	1×
Energy per kg	–70% vs. vertical	High (static LEDs)	Solar · free
Water usage	–95% closed loop	–70% to –80%	Very high
Pharma / GMP	• Ready	○ Retrofit required	○ Not achievable
Pesticide-free output	• Guaranteed	• Mostly	○ Rarely
Scale unit	1 Loop → gigafarm	Whole facility	Hectare blocks

Break-even FY28. €205M revenue by FY32.

\$ NDA-GATED



THE ASK

Series A • open to multiple instruments

SERIES A • SCALING ROUND → SERIES B

€2M

Series A tranche to deliver our first 10 commercial Loop installations and prepare the Series B growth round in FY27.

TOTAL FUNDING THROUGH SERIES B: €6M

INSTRUMENTS \U2014 OPEN

- Direct Equity
- SAFE
- Convertible
- tokenize.it

Structure tailored to investor preference.

USE OF FUNDS

45% Commercial roll-out
First 10 Loop installations

25% R&D • protocols
22 → 50+ grow recipes

20% Team & hires
Engineering • ops • sales

10% IP & compliance
Patent maintenance • GMP cert.

THE TEAM

Operators who've built and scaled before.

**Jan Bredack****FOUNDER · CEO**jan.bredack@orbi.farm

Founder, OrbiFarm®. Serial entrepreneur in sustainable consumer & deep-tech. Managing Director, OrbiFarm GmbH.

**Dr. Jan-Gerd Frerichs****CTO**jan-gerd.frerichs@orbi.farm

Technology lead. Drives automation, mechatronics and platform architecture across the OrbiFarm® Loop system.

**Maja Bredack****CO-FOUNDER · COO**maja.bredack@orbi.farm

Operations & organisational leadership. Scales OrbiFarm® from pilot to fleet — finance, processes, people.

**Lara Pook****HEAD OF CONSTRUCTION & DEVELOPMENT**lara.pook@orbi.farm

Leads hardware engineering and deployment. Owns Loop system design, build-out and site roll-out.

Let's grow, everywhere.

Join the Series A and secure your position in the most capital-efficient TCEA platform on the planet.

FOUNDER

Jan Bredack

CTO

Dr. Jan-Gerd Frerichs

WEB

orbi.farm

INVEST

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